

Case 1 It's Global Warming, Stupid

By Paul M. Barrett on November 01, 2012

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Hurricane Sandy churns off the coast of Florida as a line of clouds associated with a powerful cold front approaches the U.S. East Coast on Oct. 26, 2012

Yes, yes, it's unsophisticated to blame any given storm on climate change. Men and women in white lab coats tell us—and they're right—that many factors contribute to each severe weather episode. Climate deniers exploit scientific complexity to avoid any discussion at all.

Clarity, however, is not beyond reach. Hurricane Sandy demands it: At least 40 U.S. deaths. Economic losses expected to climb as high as \$50 billion. Eight million homes without power. Hundreds of thousands of people evacuated. More than 15,000 flights grounded. Factories, stores, and hospitals shut. Lower Manhattan dark, silent, and underwater.

An unscientific survey of the social networking literature on Sandy reveals an illuminating tweet (you read that correctly) from Jonathan Foley, director of the Institute on the Environment at the University of Minnesota. On Oct. 29, Foley thumbed thusly: "Would this kind of storm happen without climate change? Yes. Fueled by many factors. Is storm stronger because of climate

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change? Yes.” Eric Pooley, senior vice president of the Environmental Defense Fund (and former deputy editor of Bloomberg Businessweek), offers a baseball analogy: “We can’t say that steroids caused any one home run by Barry Bonds, but steroids sure helped him hit more and hit them farther. Now we have weather on steroids.”

In an Oct. 30 blog post, Mark Fischetti of Scientific American took a spin through Ph.D.-land and found more and more credentialed experts willing to shrug off the climate caveats. The broadening consensus: “Climate change amps up other basic factors that contribute to big storms. For example, the oceans have warmed, providing more energy for storms. And the Earth’s atmosphere has warmed, so it retains more moisture, which is drawn into storms and is then dumped on us.” Even those of us who are science-phobic can get the gist of that.

Sandy featured a scary extra twist implicating climate change. An Atlantic hurricane moving up the East Coast crashed into cold air dipping south from Canada. The collision supercharged the storm’s energy level and extended its geographical reach. Pushing that cold air south was an atmospheric pattern, known as a blocking high, above the Arctic Ocean. Climate scientists Charles Greene and Bruce Monger of Cornell University, writing earlier this year in *Oceanography*, provided evidence that Arctic icemelts linked to global warming contribute to the very atmospheric pattern that sent the frigid burst down across Canada and the eastern U.S.

If all that doesn’t impress, forget the scientists ostensibly devoted to advancing knowledge and saving lives. Listen instead to corporate insurers committed to compiling statistics for profit. On Oct. 17 the giant German reinsurance company Munich Re issued a prescient report titled *Severe Weather in North America*. Globally, the rate of extreme weather events is rising, and “nowhere in the world is the rising number of natural catastrophes more evident than in North America.” From 1980 through 2011, weather disasters caused losses totaling \$1.06 trillion. Munich Re found “a nearly quintupled number of weather-related loss events in North America for the past three decades.” By contrast, there was “an increase factor of 4 in Asia, 2.5 in Africa, 2 in Europe, and 1.5 in South America.” Human-caused climate change “is believed to contribute to this trend,” the report said, “though it influences various perils in different ways.”

Global warming “particularly affects formation of heat waves, droughts, intense precipitation events, and in the long run most probably also tropical cyclone intensity,” Munich Re said. This July was the hottest month recorded in the U.S. since record-keeping began in 1895, according to the National Oceanic and Atmospheric Administration. The U.S. Drought Monitor reported that two-thirds of the continental U.S. suffered drought conditions this summer.

Granted, Munich Re wants to sell more reinsurance (backup policies purchased by other insurance companies), so maybe it has a selfish reason to stir anxiety. But it has no obvious motive for fingering global warming vs. other causes. “If the first effects of climate change are already perceptible,” said Peter Hoppe, the company’s chief of geo-risks research, “all alerts and measures against it have become even more pressing.”

Which raises the question of what alerts and measures to undertake. In his book *The Conundrum*, David Owen, a staff writer at the *New Yorker*, contends that as long as the West places high and unquestioning value on economic growth and consumer gratification—with China and the rest of the developing world right behind—we will continue to burn the fossil fuels whose emissions trap heat in the atmosphere. Fast trains, hybrid cars, compact fluorescent light bulbs, carbon offsets—they're just not enough, Owen writes.

Yet even he would surely agree that the only responsible first step is to put climate change back on the table for discussion. The issue was MIA during the presidential debates and, regardless of who wins on Nov. 6, is unlikely to appear on the near-term congressional calendar. After Sandy, that seems insane.

Mitt Romney has gone from being a supporter years ago of clean energy and emission caps to, more recently, a climate agnostic. On Aug. 30, he belittled his opponent's vow to arrest climate change, made during the 2008 presidential campaign. "President Obama promised to begin to slow the rise of the oceans and heal the planet," Romney told the Republican National Convention in storm-tossed Tampa. "My promise is to help you and your family." Two months later, in the wake of Sandy, submerged families in New Jersey and New York urgently needed some help dealing with that rising-ocean stuff.

Obama and his strategists clearly decided that in a tight race during fragile economic times, he should compete with Romney by promising to mine more coal and drill more oil. On the campaign trail, when Obama refers to the environment, he does so only in the context of spurring "green jobs." During his time in office, Obama has made modest progress on climate issues. His administration's fuel-efficiency standards will reduce by half the amount of greenhouse gas emissions from new cars and trucks by 2025. His regulations and proposed rules to curb mercury, carbon, and other emissions from coal-fired power plants are forcing utilities to retire some of the dirtiest old facilities. And the country has doubled the generation of energy from renewable sources such as solar and wind.

Still, renewable energy accounts for less than 15 percent of the country's electricity. The U.S. cannot shake its fossil fuel addiction by going cold turkey. Offices and factories can't function in the dark. Shippers and drivers and air travelers will not abandon petroleum overnight. While scientists and entrepreneurs search for breakthrough technologies, the next president should push an energy plan that exploits plentiful domestic natural gas supplies. Burned for power, gas emits about half as much carbon as coal. That's a trade-off already under way, and it's worth expanding. Environmentalists taking a hard no-gas line are making a mistake.

Conservatives champion market forces—as do smart liberals—and financial incentives should be part of the climate agenda. In 2009 the House of Representatives passed cap-and-trade legislation

that would have rewarded more nimble industrial players that figure out how to use cleaner energy. The bill died in the Senate in 2010, a victim of Tea Party-inspired Republican obstructionism and Obama's decision to spend his political capital to push health-care reform.

Despite Republican fanaticism about all forms of government intervention in the economy, the idea of pricing carbon must remain a part of the national debate. One politically plausible way to tax carbon emissions is to transfer the revenue to individuals. Alaska, which pays dividends to its citizens from royalties imposed on oil companies, could provide inspiration (just as Romneycare in Massachusetts pointed the way to Obamacare).

Ultimately, the global warming crisis will require global solutions. Washington can become a credible advocate for moving the Chinese and Indian economies away from coal and toward alternatives only if the U.S. takes concerted political action. At the last United Nations conference on climate change in Durban, South Africa, the world's governments agreed to seek a new legal agreement that binds signatories to reduce their carbon emissions. Negotiators agreed to come up with a new treaty by 2015, to be put in place by 2020. To work, the treaty will need to include a way to penalize countries that don't meet emission-reduction targets—something the U.S. has until now refused to support.

If Hurricane Sandy does nothing else, it should suggest that we need to commit more to disaster preparation and response. As with climate change, Romney has displayed an alarmingly cavalier attitude on weather emergencies. During one Republican primary debate last year, he was asked point-blank whether the functions of the Federal Emergency Management Agency ought to be turned back to the states. "Absolutely," he replied. Let the states fend for themselves or, better yet, put the private sector in charge. Pay-as-you-go rooftop rescue service may appeal to plutocrats; when the flood waters are rising, ordinary folks welcome the National Guard. It's possible Romney's kill-FEMA remark was merely a pander to the Right, rather than a serious policy proposal. Still, the reconfirmed need for strong federal disaster capability—FEMA and Obama got glowing reviews from New Jersey Governor Chris Christie, a Romney supporter—makes the Republican presidential candidate's campaign-trail statement all the more reprehensible.

The U.S. has allowed transportation and other infrastructure to grow obsolete and deteriorate, which poses a threat not just to public safety but also to the nation's economic health. With once-in-a-century floods now occurring every few years, New York Governor Andrew Cuomo and New York City Mayor Michael Bloomberg said the country's biggest city will need to consider building surge protectors and somehow waterproofing its enormous subway system. "It's not prudent to sit here and say it's not going to happen again," Cuomo said. "I believe it is going to happen again."

David Rothkopf, the chief executive and editor-at-large of Foreign Policy, noted in an Oct. 29 blog post that Sandy also brought his hometown, Washington, to a standstill, impeding affairs of

state. To lessen future impact, he suggested burying urban and suburban power lines, an expensive but sensible improvement.

Where to get the money? Rothkopf proposed shifting funds from post-Sept. 11 bureaucratic leviathans such as the Department of Homeland Security, which he alleges is shot through with waste. In truth, what's lacking in America's approach to climate change is not the resources to act but the political will to do so. A Pew Research Center poll conducted in October found that two-thirds of Americans say there is "solid evidence" the earth is getting warmer. That's down 10 points since 2006. Among Republicans, more than half say it's either not a serious problem or not a problem at all.

Such numbers reflect the success of climate deniers in framing action on global warming as inimical to economic growth. This is both shortsighted and dangerous. The U.S. can't afford regular Sandy-size disruptions in economic activity. To limit the costs of climate-related disasters, both politicians and the public need to accept how much they're helping to cause them.

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Analysis

1. Identify what a "hurricane" influences and what influences it. Quantify the variables identified.
2. Develop a reinforcing loop.
3. Create a balancing loop. Evaluate its significance.